

Ricky Suarez

Grazioso Salvare Rescue Dog Dashboard

Description

The Grazioso Salvare Rescue Dog Dashboard is a full stack client server application designed to help an international rescue animal training company identify dogs suitable for specialized search and rescue training. The application connects to a MongoDB database containing Austin Animal Center outcomes data and provides an interactive dashboard that allows users to filter, visualize, and explore potential rescue dog candidates.

The dashboard applies database concepts, client side interaction, and the MVC design pattern to provide a user friendly decision support system. Users can filter animals by rescue specialization and dynamically view updates through a data table, geolocation map, and outcome frequency chart.

Features

- MongoDB database integration using a custom Python CRUD module
- Interactive dashboard built with Dash and JupyterDash
- Radio button filtering for rescue categories
- Dynamic data table with sorting, filtering, and pagination
- Geolocation map visualization using Leaflet
- Outcome type frequency chart for data insights
- Reset functionality to restore unfiltered dataset
- MVC architecture separating model, view, and controller logic

Rescue Filter Categories

The dashboard supports filtering dogs suitable for the following rescue types:

- Water Rescue
- Wilderness Rescue
- Disaster or Individual Tracking
- Reset to view all animals

Each filter dynamically updates the table, map, and chart widgets.

Installation

Requirements

- Python 3.x
- MongoDB running locally
- Jupyter Lab environment
- Required Python libraries:

`pip install pandas numpy matplotlib plotly dash jupyter-dash dash-leaflet pymongo`

Usage

Step 1

Import the Austin Animal Center dataset into MongoDB and create the animals collection.

Step 2

Run the CRUD Python module to verify database connectivity.

Step 3

Launch the dashboard notebook:

Run all cells in `ProjectTwoDashboard.ipynb`

Step 4

Interact with the dashboard using radio filters to explore rescue candidate dogs.

Screenshots

Include the following screenshots in the repository:

- Initial dashboard state with all widgets visible
- Water rescue filter applied
- Wilderness rescue filter applied
- Disaster or tracking filter applied
- Reset filter returning to original dataset

MVC Architecture

Model

MongoDB database storing Austin Animal Center outcomes data

View

Dashboard widgets including data table, map, and chart

Controller

Python CRUD module and Dash callbacks handling filtering and visualization logic

Tools and Technologies

- Python
- MongoDB
- PyMongo
- Pandas
- Plotly
- Dash and JupyterDash
- Dash Leaflet
- Jupyter Lab

Contributing

This project was created as part of an academic software engineering assignment. Contributions are not required, but suggestions for improvements are welcome.

Author

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Project Status

Completed as part of the Grazioso Salvare dashboard development project for SNHU CS 340.

License

This project is intended for educational use.

Name: ModuleSixMilestone.ipynb
Path: code_files/ModuleSixMilestone.ipynb
Last Saved: 2026-02-22 00:00
Last Checkpoint: 2026-02-19 01:37

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- ☐ Water Rescue
- ☐ Wilderness Rescue
- ☐ Disaster / Tracking
- ☒ Reset

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color
	filter					
☒	3	2 years	A716330	Dog	Chihuahua Shorthair Mix	Brown/White
☐	9	3 years	A720214	Dog	Labrador Retriever Mix	Red/White
☐	10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie
☐	8	1 year	A736551	Dog	Labrador Retriever/Australian Cattle Dog	Black
☐	11	1 year	A721199	Dog	Dachshund Wirehair Mix	Tan/White
☐	12	1 year	A664843	Dog	Pit Bull Mix	Brown/White
☐	13	1 year	A700408	Cat	Domestic Shorthair Mix	Brown/White
☐	14	2 years	A742287	Dog	Boxer/Bullmastiff	Brown Brindle/White
☐	15	5 years	A722742	Dog	Mix of German Shepherd and Labrador Retriever	Black/White

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- ☒ Water Rescue
☐ Wilderness Rescue
☐ Disaster / Tracking
☐ Reset

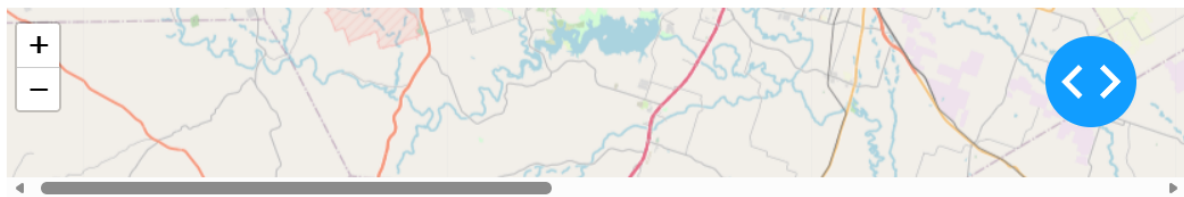
	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	da
	filter							
☒	36	6 months	A706953	Dog	Labrador Retriever Mix	Yellow	2014-12-06	2015-07-06 11
☐	732	2 years	A749782	Dog	Labrador Retriever Mix	Tan/White	2015-05-19	2017-07-25 14
☐	1121	1 year	A757158	Dog	Labrador Retriever Mix	White/Black	2016-08-30	2017-08-31 14
☐	1628	9 months	A740471	Dog	Labrador Retriever Mix	Tan/White	2016-03-17	2016-12-23 17
☐	1757	7 months	A742767	Dog	Labrador Retriever Mix	Black	2016-06-27	2017-02-14 15
☐	1988	1 year	A762781	Dog	Labrador Retriever Mix	Black/White	2016-11-27	2017-02-03 13
☐	2041	2 years	A702745	Dog	Labrador Retriever Mix	Black	2013-05-22	2017-02-22 11
☐	2225	2 years	A757341	Dog	Labrador Retriever Mix	Black/White	2015-09-01	2017-10-03 12
☐	2310	6 months	A697748	Dog	Labrador Retriever Mix	Yellow	2017-12-06	2018-06-06 17

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- ☐ Water Rescue
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	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime
	filter							
☒	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54:00
☐	6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23:00
☐	3130	2 years	A721834	Dog	Siberian Husky	Brown/White	2014-03-05	2016-03-23 16:23:00
☐	6191	2 years	A704101	Dog	Siberian Husky	Black/White	2013-06-01	2015-06-02 16:41:00

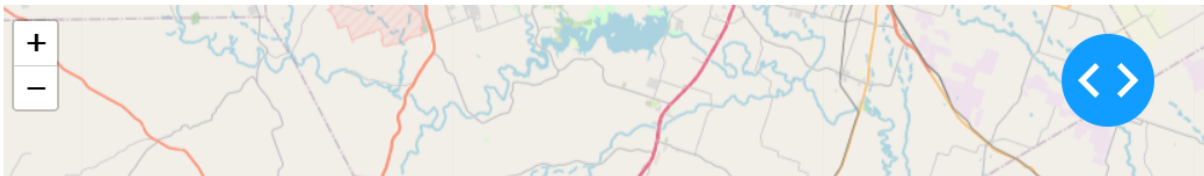


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	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime
	filter							
☒	3767	4 years	A712291	Dog	Bloodhound	Red	2011-09-20	2015-09-22 15:43:00
☐	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54:00
☐	2987	4 years	A694614	Dog	Rottweiler	Black/Brown	2011-01-01	2015-01-01 14:25:00
☐	6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23:00



All Queries display a bar graph and map

